

Image Processing and Computer Vision

Broken Biscuit Detection using Classical Image Processing Techniques

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Assignment 02



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1 Introduction

In this assignment, you will develop an image processing solution to detect whether biscuits are **broken or intact**. The objective is to apply **classical image processing techniques** to analyze real-world images and perform automatic detection.

You must collect your own dataset by capturing images of biscuits arranged on A4 sheets. Your program should detect each biscuit in the image and determine whether it is broken or intact.

Important: Machine Learning or Deep Learning methods are **not allowed**. You must use only **classical image processing techniques** such as thresholding, edge detection, contour detection, and morphological operations.

2 Dataset Preparation

You must create your own dataset by capturing images of biscuits.

2.1 Biscuit Types

You must use two types of biscuits:

- Circular biscuits
- Square biscuits

Place **circular biscuits on one white A4 sheet** and **square biscuits on a separate white A4 sheet**.

Your dataset must contain a **mix of intact and broken biscuits**.

2.2 Image Capture Instructions

Follow these instructions when capturing images:

- Place biscuits on a **white A4 sheet**.
- Capture images from a **top-down view (approximately 90°)**.
- Each image should contain **multiple biscuits**.
- Ensure there is a **mix of intact and broken biscuits**.
- Ensure **good lighting** and avoid strong shadows.
- Capture images by **changing the arrangement of biscuits each time**.

2.3 Sample Images

Figure 1 shows an example of circular biscuits, and Figure 2 shows an example of square biscuits.

You must capture **at least 10 images**.

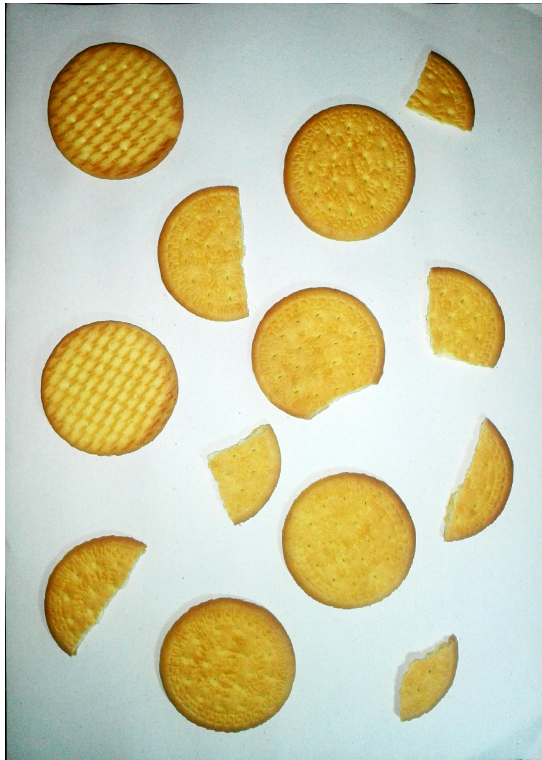


Figure 1: Sample circular biscuits

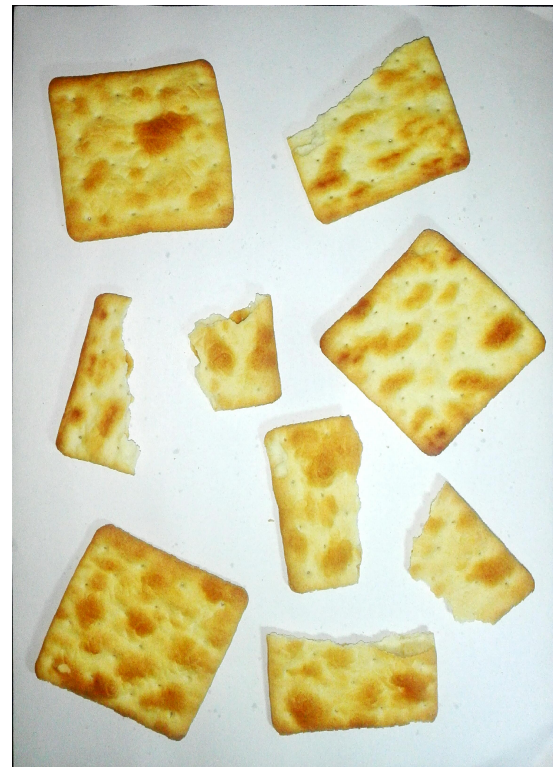


Figure 2: Sample square biscuits

3 Image Processing Task

You must develop a program that performs the following tasks:

1. Detect each biscuit in the image.
2. Determine whether each biscuit is **broken** or **intact**.
3. Annotate the image by marking each detected biscuit.
4. Display a label on the image indicating the detection result.

Example labels may include:

- Intact Biscuit
- Broken Biscuit

Each biscuit detected in the image must be clearly labeled in the output.

4 Allowed Methods

You must use only **classical image processing techniques**. Examples include:

- Image preprocessing
- Grayscale conversion

- Thresholding
- Edge detection
- Contour detection
- Morphological operations

Machine Learning or Deep Learning methods are strictly prohibited.

5 GitHub Repository Requirement

You must create a **GitHub repository** for this assignment and upload your project files.

The repository must contain:

- Source code
- Captured input images
- Annotated output images
- README.md file

5.1 Repository Guidelines

- Organize files using a clear folder structure.
- Use **meaningful commit messages**.
- Show the development process through multiple commits.

5.2 README File

Your README file must include:

- Project title
- Problem description
- Tools and libraries used
- Image processing methods used
- Instructions to run the code
- Example output images

6 Report

You must prepare a **report (PDF)** describing your work.

The report must include the following sections:

- Introduction
- Dataset Description
- Methodology
- Implementation
- Results
- Discussion
- Conclusion
- GitHub Repository Link

The **GitHub repository link must be included in the report.**

7 Submission

You must submit **only the report (PDF)**.

The report file must be named using the following format:

`Assignment_02_REG_NO.pdf`

Note: Replace YOUR_REG_NO with your actual registration number. Submit only the ZIP file.

If you record a video of biscuits moving on a conveyor belt and apply your program to **detect and track** each biscuit, you can earn extra marks for the module. You must annotate the video with labels such as **Intact Biscuit** and **Broken Biscuit** for each detected biscuit. Include the video in your GitHub repository and provide the video link in your report. **This is optional and not necessary to complete the assignment.** You can achieve full marks without this task.

8 IMPORTANT

Program should not copied from any source. It should be the original work of the student. Plagiarism will not be tolerated and will result in a zero grade for the assignment.